

Introduction to Machine Learning and AI in Clinical Bioinformatics

Instructor: Caryn Geady, PhD

Date: 12th Jan 2026

Today's Objectives

- Course frontmatter:
 - Course Outline / Critical Path
 - Road-map;
 -  ○ Expectations;
- Key concepts and terminology:
 - Machine learning and AI;
 - Supervised vs. unsupervised learning;
- Select applications in clinical bioinformatics:
 - Multi-omics data;
 - Biomarker discovery;
- Introduction to GitHub



Course Frontmatter & Logistics

Course Roadmap

- **Frequently-used algorithms in biomedical science:**

- Supervised Learning;
- Unsupervised Learning;



- **Applications:**

- Survival Analysis
- Radiomics;
- Pharmacogenomics;
- Open Science / Multi-omics

- **Assessments:**

- 1 X participation/GitHub
- 4 X assignments;
- 1 X midterm;
- 1 X capstone project/presentation/report



SVM



KNN



Regression



Random forest



Survival analysis



Meta-analysis



Hierarchical



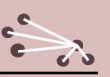
K-means



PCA



tSNE



- Cancer vs healthy classification (gene expression)

- Multiclass tissue classification (gene expression)

- Genome-wide association analysis (SNP)

- Pathway-based classification (gene expression, SNP)

- Modeling time-to-event outcomes (overall survival)

- Combining results from multiple studies (biomarker discovery)

- Protein family clustering (amino acid sequence)

- Clustering genes by chromosomes (gene expression)

- Classification of outliers (gene expression)

- Data visualization (single cell RNA-sequencing)

Course Expectations

- **Understand core ML techniques:**
 - Supervised learning to unsupervised learning and beyond;
 - Matching the correct technique for the specific clinical bioinformatics problem;
- **Develop a practical coding toolbox:**
 - Build a personalized repository of reusable code snippets for common ML tasks;
 - Include solutions for data processing, model training, evaluation and visualization;
- **Bridge theory with application:**
 - Learn the “why” behind ML techniques, as well as the “how” to implement them in Python;
 - Translate theoretical concepts into real-world workflows for clinical bioinformatics.



Housekeeping / Important Resources

- Humber's Statement on AI:
<https://humber.ca/academic-division/academic-integrity/statement-on-ai>
- Use of AI in This Course:
 - AI tools (e.g., Copilot, ChatGPT) are allowed and expected;
 - Transparency about AI usage is required in all assignments;
 - You must still understand and explain your own work!
- See Blackboard for Course Outline and Critical Path.

Assessment Breakdown

Assessment	Weight	Deliverable(s)
GitHub/Participation	10%	● GitHub repositories ● Keep showing up!
Assignments	40% (4 X 10%)	● Code ● Short answer/result
Midterm	15%	● Multiple choice
Capstone Project	35%	● Short report ● Presentation

Lab Logistics

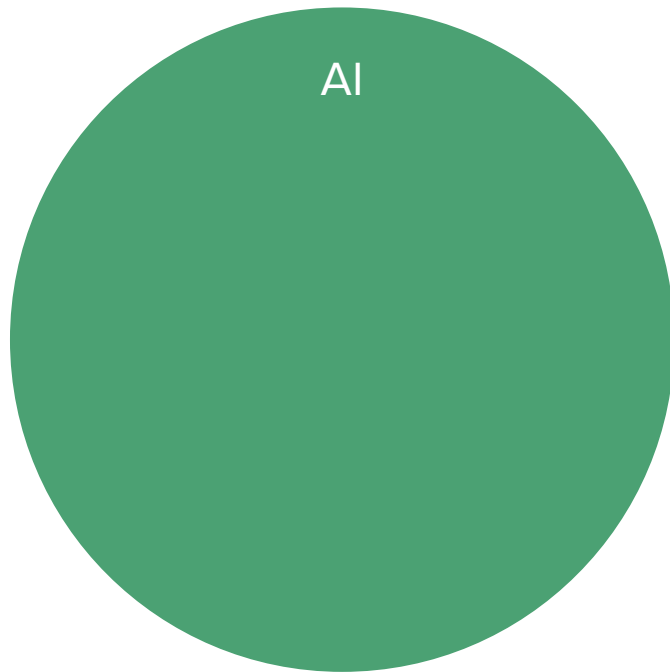
- **Day of Week:** Thursday
- **Format:** .ipynb notebook skeleton with questions
- **Schedule*:**

Time	Activity
6:05pm - 6:15pm	● Q&A
6:15pm - 6:45pm	● Code Walkthrough
6:45pm - 7:30pm	● Group Work
7:30pm - 7:40pm	● Break
7:40pm - 8:40pm	● Group Discussion

Key Concepts and Terminology

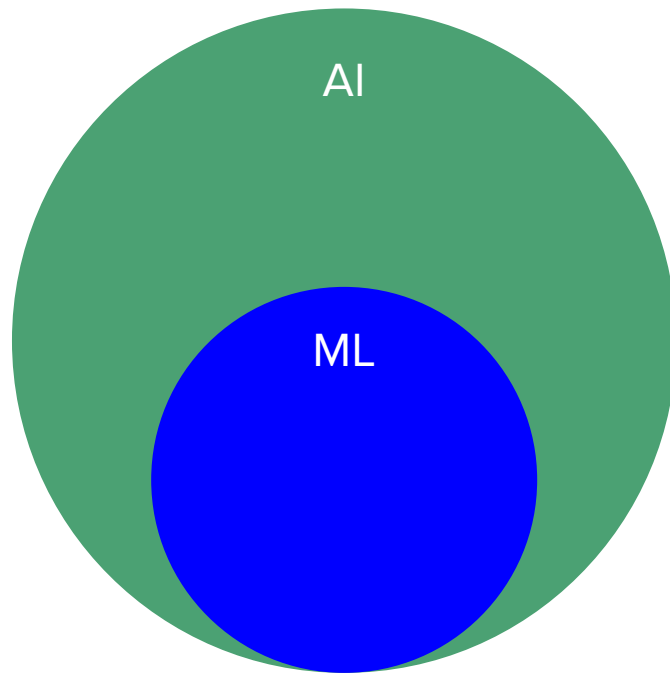
What is Artificial Intelligence (AI)?

- Simulation of human intelligence in machines;
- Spans various fields, from natural language processing to robotics;
- Includes any system that can mimic human-like tasks (i.e., decision-making, pattern recognition)



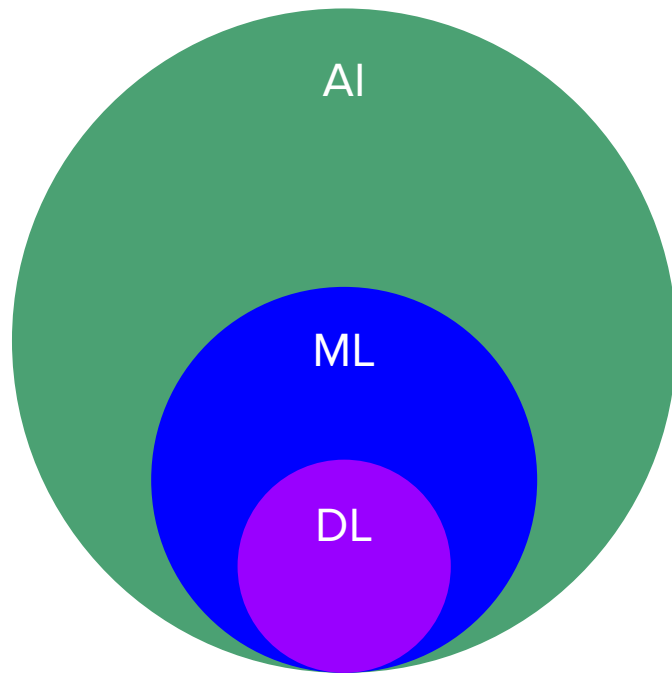
What is Machine Learning (ML)?

- Subset of AI that uses data to “train” systems for prediction or classification;
- Ubiquitously used in modern society; 

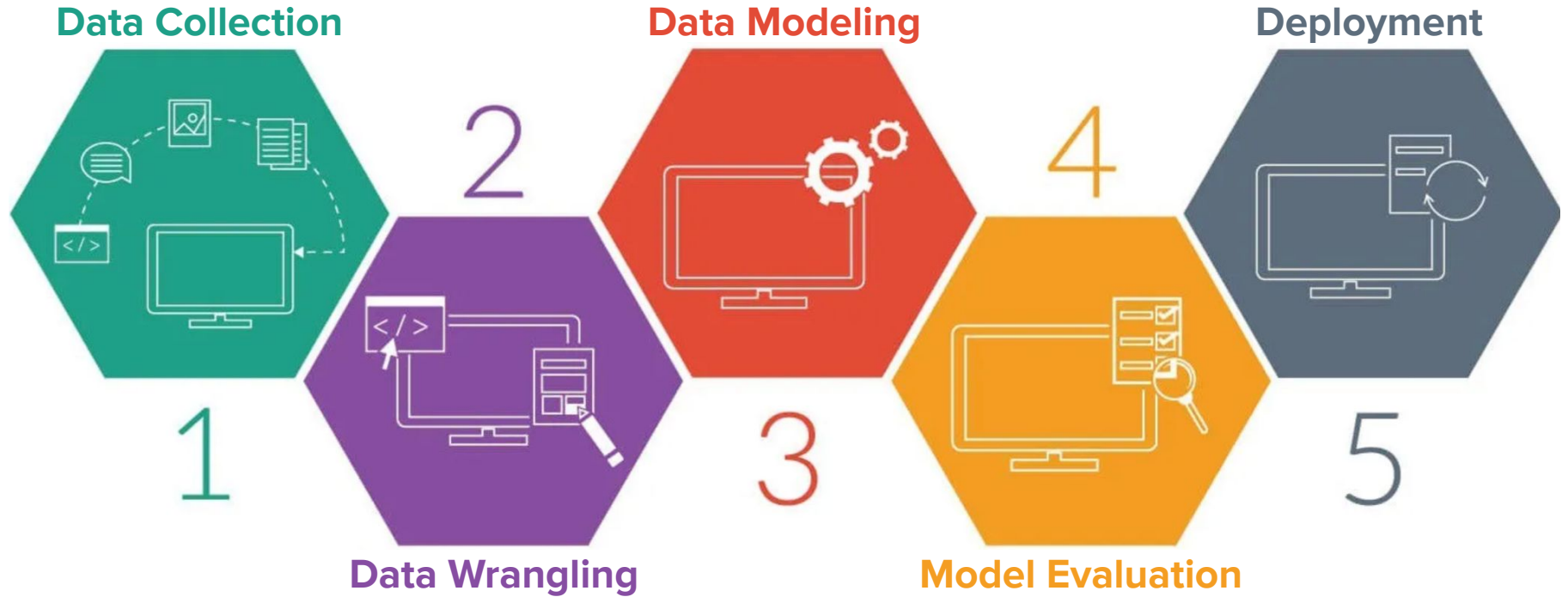


What is Machine Learning (ML)?

- Subset of AI that uses data to “train” systems for prediction or classification;
- Ubiquitously used in modern society;
- Deep Learning (DL) → specialized branch of ML



AI/ML Workflow





Essential Terminology

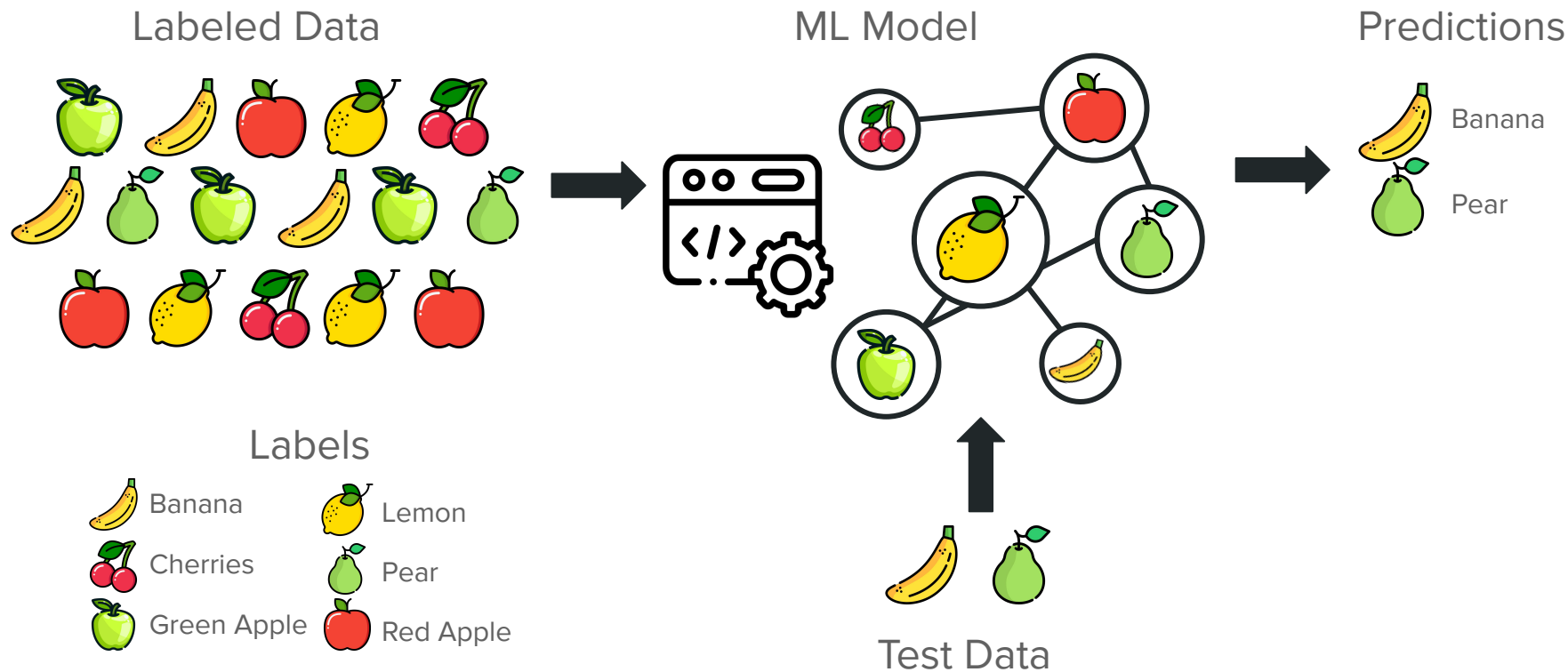
Terminology	Definition	Example(s)
Feature	Property or characteristic of the data used for analysis or modeling	Age, blood pressure, gene expression levels
Label	Target variable or outcome that a model is designed to predict	Survival, gene expression, disease state
Algorithm	A set of mathematical instructions or procedures used to create a ML model	Linear regression, decision trees, k-means clustering
Model	Output of a ML algorithm after it has been trained on data; used to make predictions or decisions	Logistic regression model predicting disease risk based on patient features



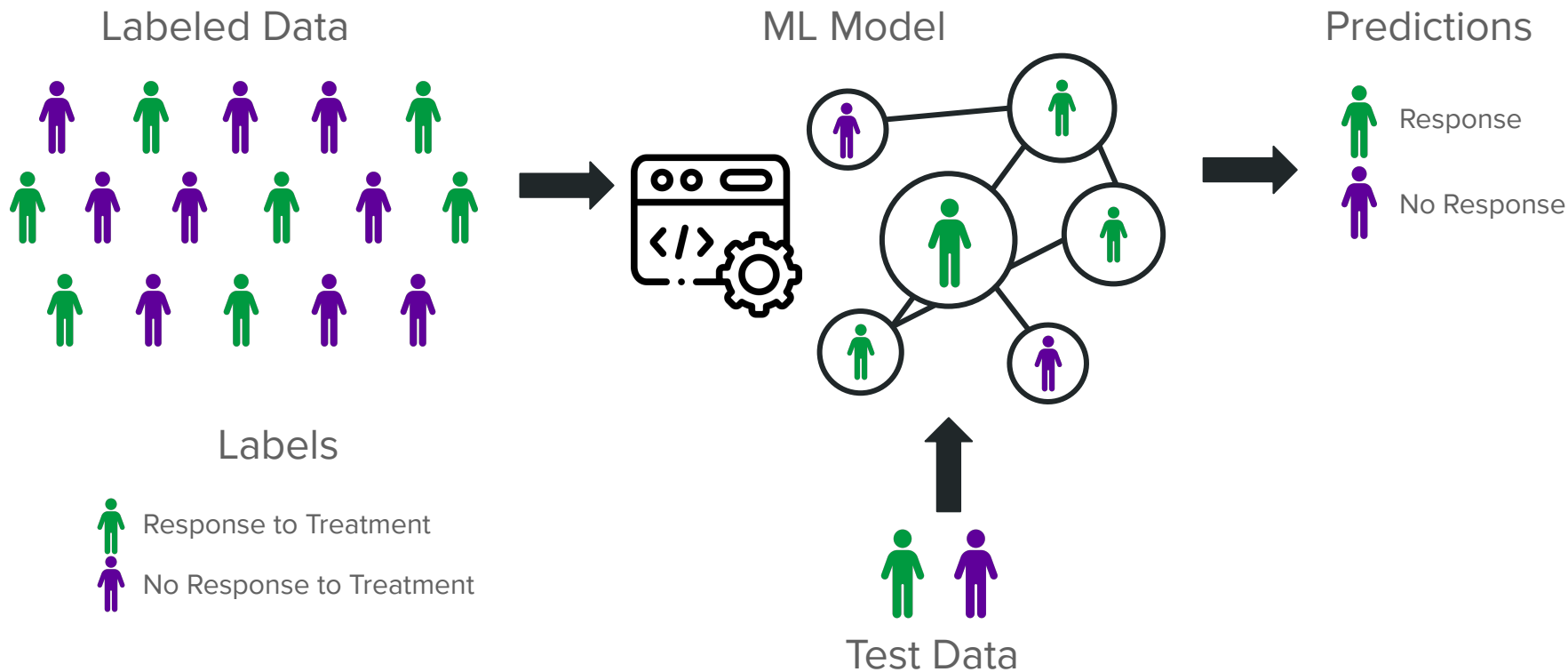
Essential Terminology (continued)

Terminology	Definition	Example(s)
Data splitting	Dividing the data into different sets for training and optimization/evaluation	80% of data for training a model, 20% of data for testing a model
Overfitting	A situation where a model learns the noise or patterns specific to the training data rather than generalizable patterns.	A model that performs perfectly on training data but fails to predict accurately on testing data.
Regularization	Techniques used to prevent overfitting by penalizing overly complex models.	Adding a penalty term to the loss function in linear regression (L1 or L2 regularization).

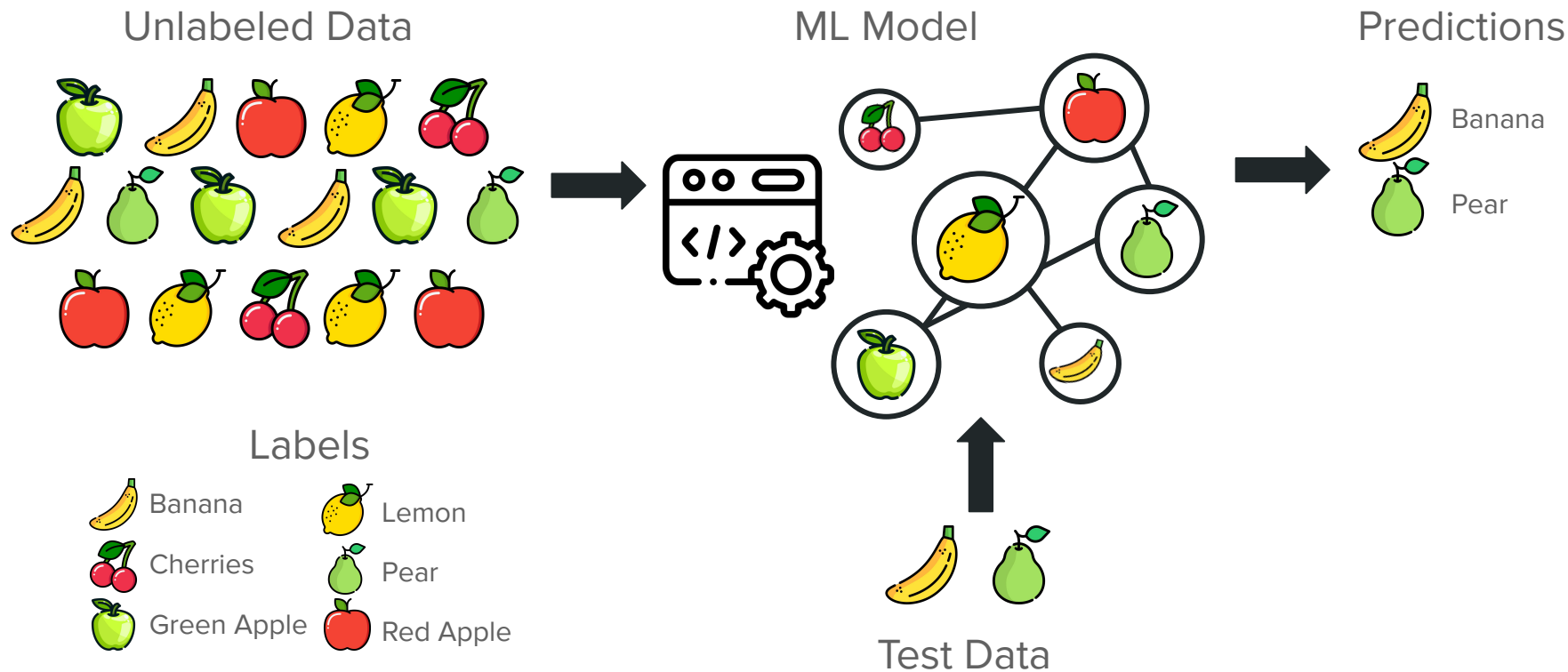
Supervised Learning: Learning from Labeled Data



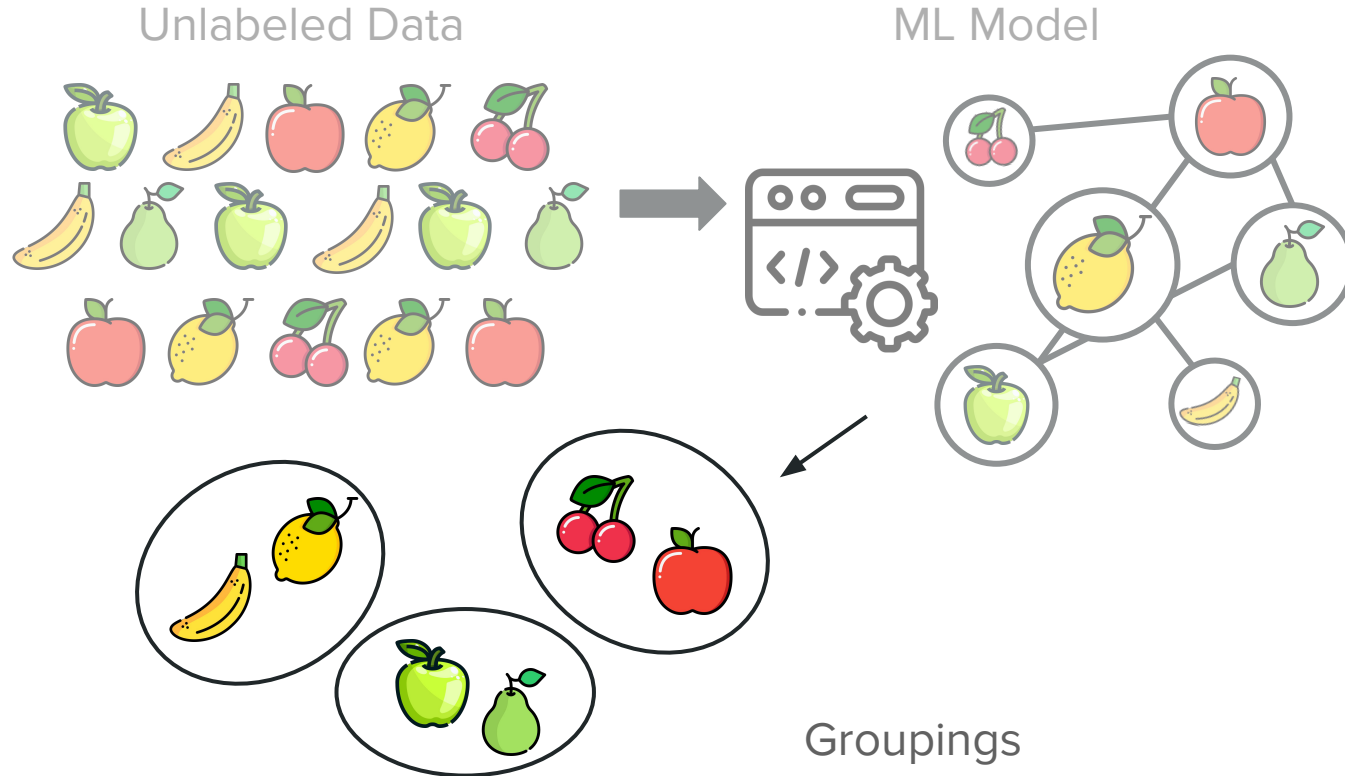
Supervised Learning: Learning from Labeled Data



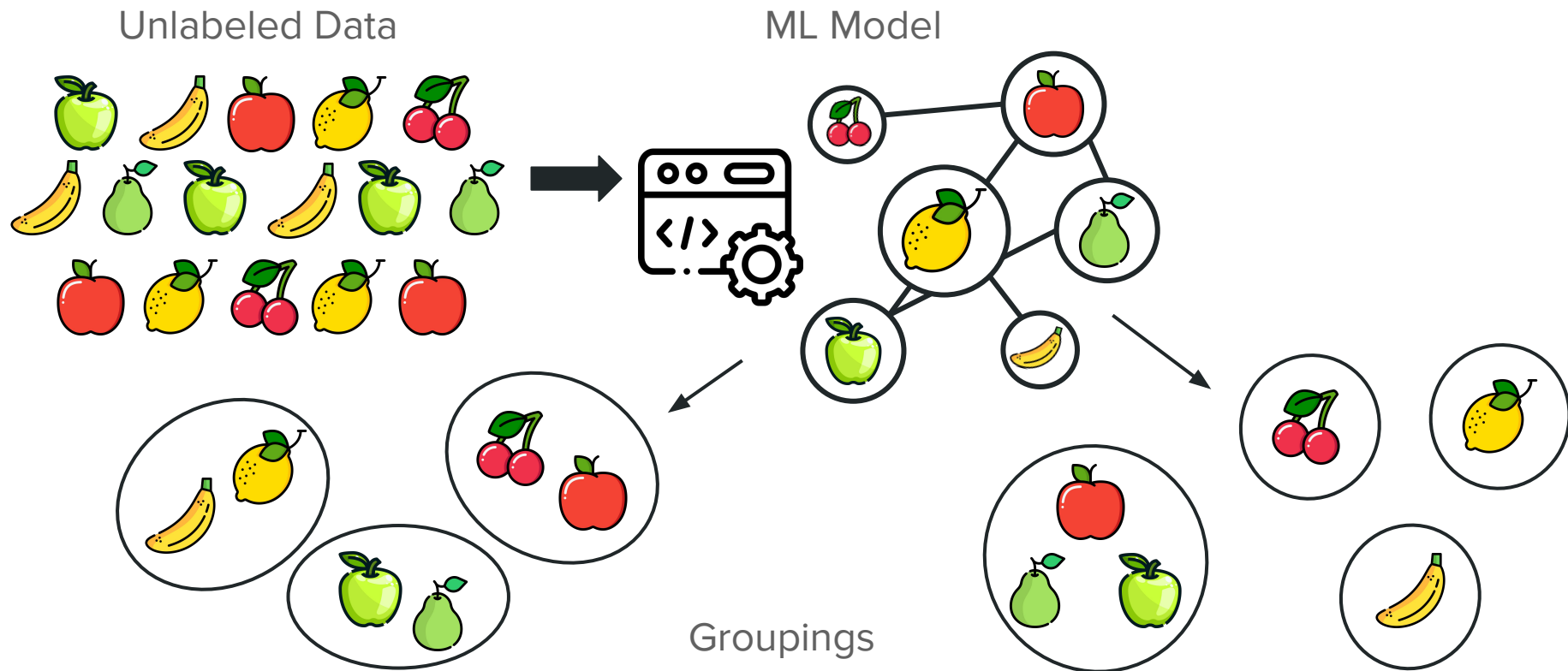
Unsupervised Learning: Finding Patterns in Data



Unsupervised Learning: Finding Patterns in Data

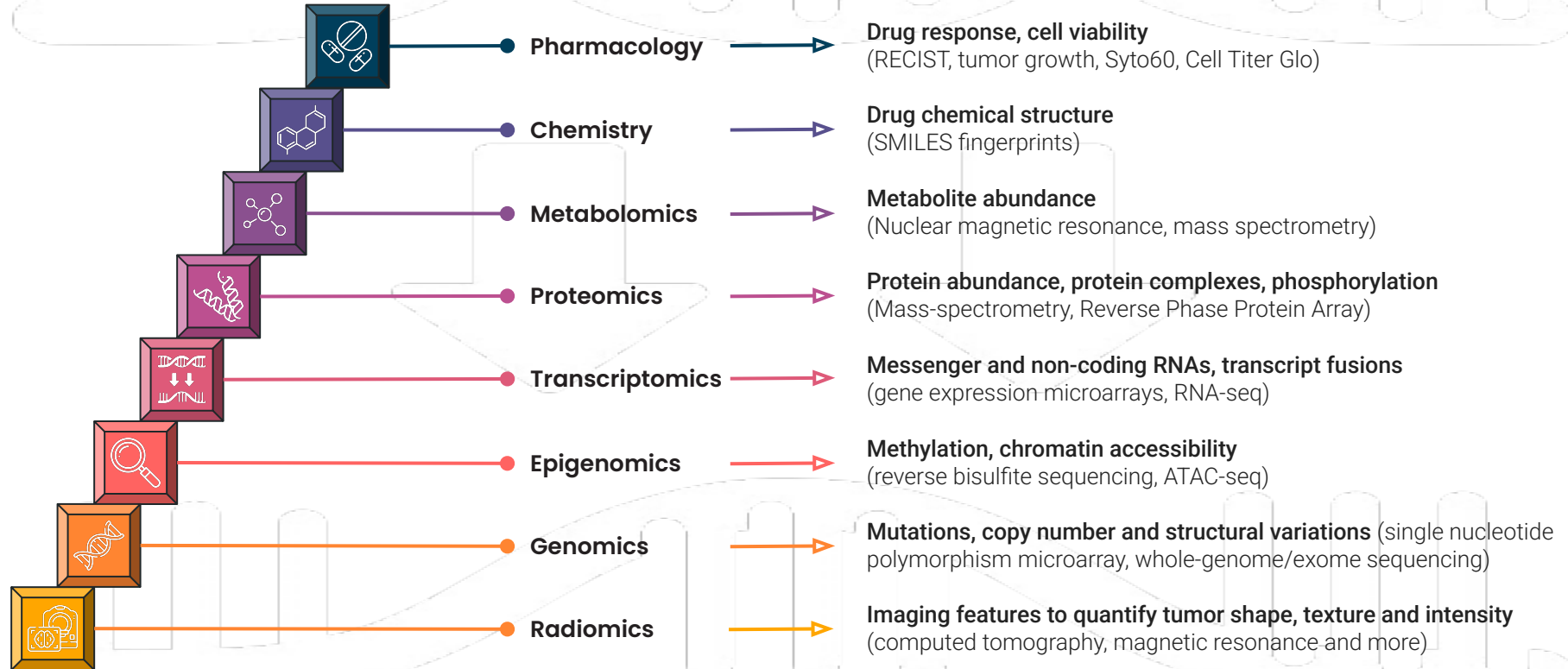


Unsupervised Learning: Finding Patterns in Data

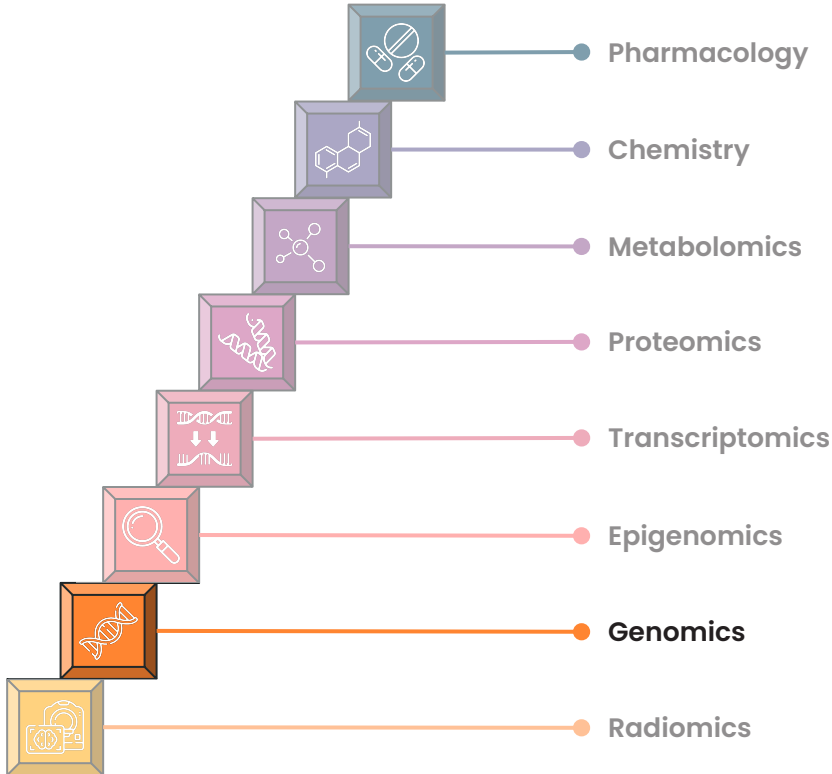


[Select] Applications in CBI

Introduction to (some) Multi-Omics Data



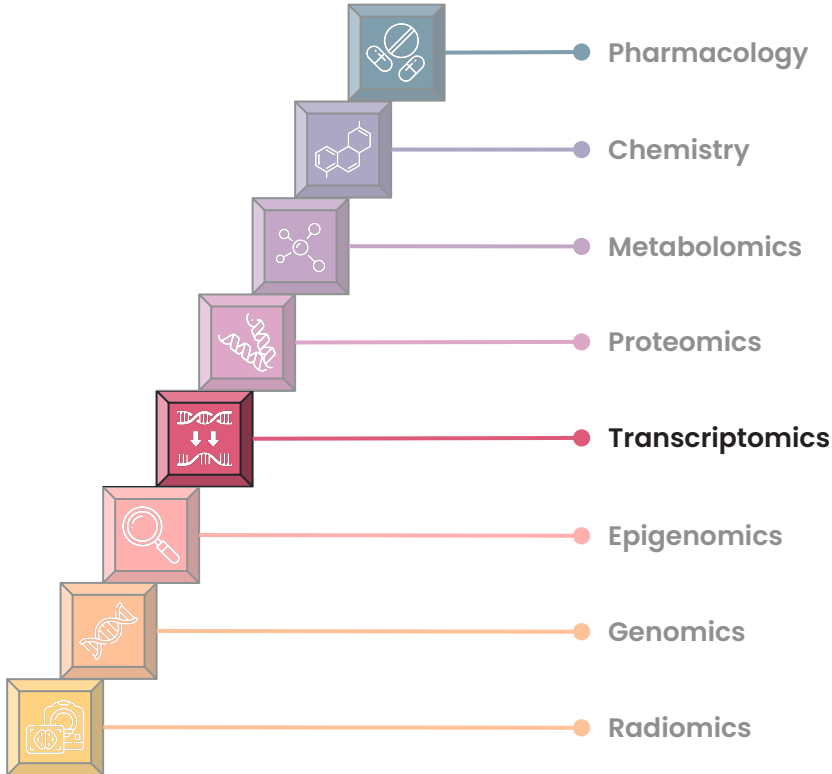
Conventional/Existing Molecular Biomarkers



Genomic Approaches:

- *BRCA1/2* mutations are biomarkers for PARP inhibitors
- *BCR-ABL1* gene fusion is a biomarker for imatinib
- *ERBB* amplification is a biomarker for ~20 drugs
- Microsatellite instability and tumor mutational burden are biomarkers for pembrolizumab

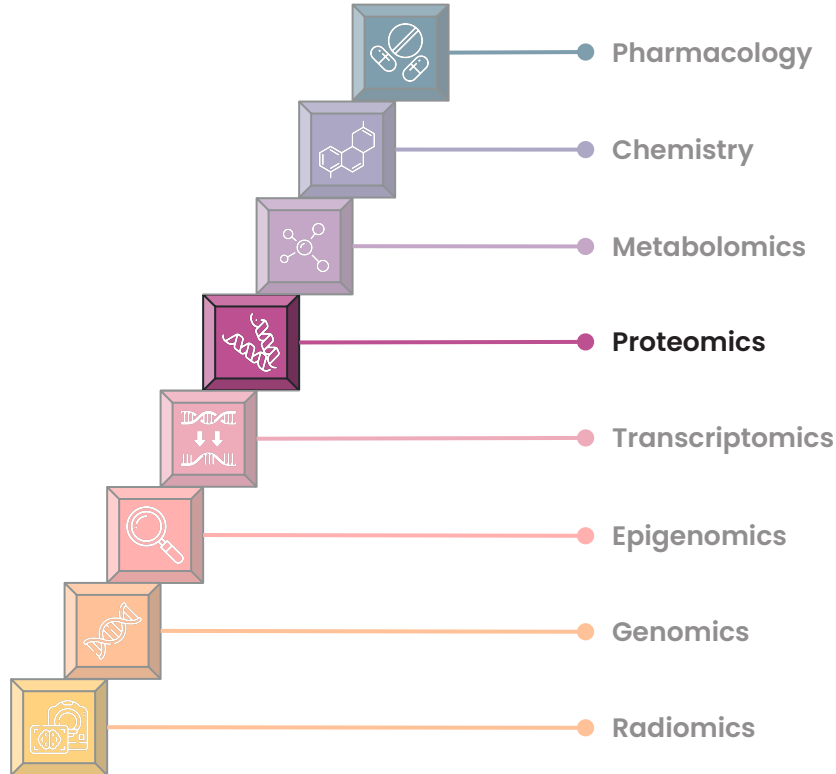
Conventional/Existing Molecular Biomarkers



Transcriptomic Approaches:

- Oncotype DX uses 21 gene expression levels to predict breast cancer recurrence risk
- PCA3 mRNA test scores predict risk for prostate cancer

Conventional/Existing Molecular Biomarkers



Proteomic Approaches:

- Carcinoembryonic Antigen (CEA) levels tested to predict colon cancer recurrence risk
- Prostate-Specific Antigen (PSA) levels tested to predict risk of prostate cancer

Medical Imaging

- Medical image segmentation → the process of delineating regions of interest in medical images, e.g. tumors or organs;
- Radiomics → extracting quantitative features from medical images to characterize disease phenotypes

Article | Published: 07 December 2020

nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation

[Fabian Isensee](#), [Paul F. Jaeger](#), [Simon A. A. Kohl](#), [Jens Petersen](#) & [Klaus H. Maier-Hein](#) 

[Nature Methods](#) **18**, 203–211 (2021) | [Cite this article](#)

125k Accesses | **3377** Citations | **254** Altmetric | [Metrics](#)

Article | [Open access](#) | Published: 03 June 2014

Decoding tumour phenotype by noninvasive imaging using a quantitative radiomics approach

[Hugo J. W. L. Aerts](#) , [Emmanuel Rios Velazquez](#), [Ralph T. H. Leijenaar](#), [Chintan Parmar](#), [Patrick Grossmann](#), [Sara Carvalho](#), [Johan Bussink](#), [René Monshouwer](#), [Benjamin Haibe-Kains](#), [Derek Rietveld](#), [Frank Hoesers](#), [Michelle M. Rietbergen](#), [C. René Leemans](#), [Andre Dekker](#), [John Quackenbush](#), [Robert Gillies](#) & [Philippe Lambin](#)

[Nature Communications](#) **5**, Article number: 4006 (2014) | [Cite this article](#)

Clinical Decision Support Systems

- Broad utility:
 - Pre-clinical data processing;
 - Diagnosis;
 - Patient stratification;
 - Treatment decision-making;
 - Prevention and monitoring
- Challenges:
 - Lack of benchmarking and external validation;
 - Gap between research and clinical practitioners;
 - Transparency and reproducibility

Machine learning-based clinical decision support using laboratory data

Hikmet Can Çubukçu ^{1 2}, Deniz İlhan Topcu ³, Sedef Yenice ⁴

Affiliations + expand

PMID: 38015744 DOI: [10.1515/cclm-2023-1037](https://doi.org/10.1515/cclm-2023-1037)

Machine learning in clinical decision making

[Lorenz Adlung](#) ^{1 3}, [Yotam Cohen](#) ^{1 3}, [Uria Mor](#) ^{1 3}, [Eran Elinav](#) ^{1 2}  

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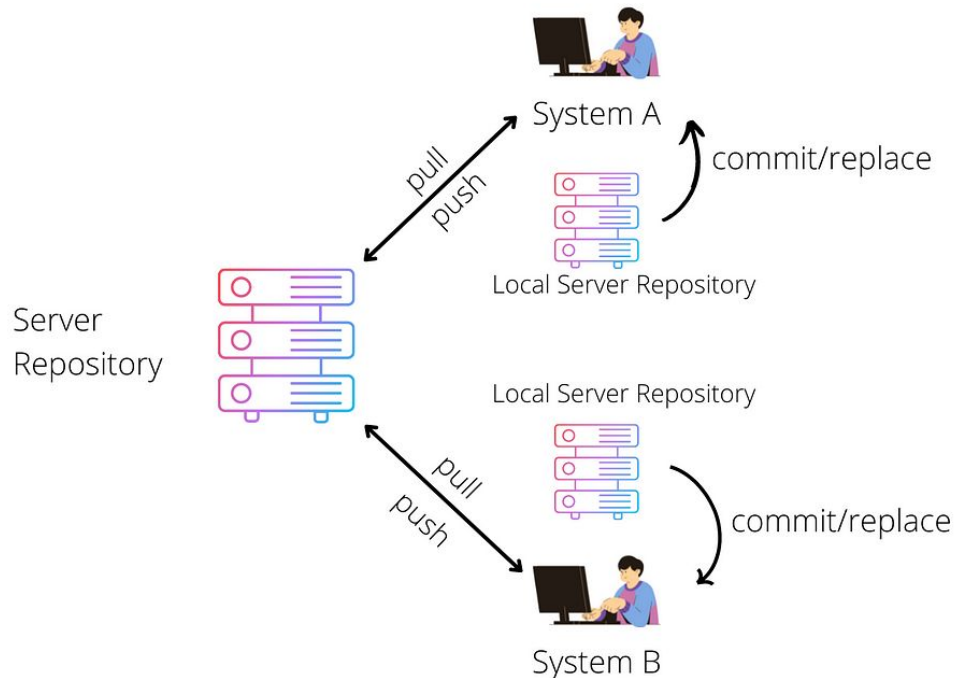
<https://doi.org/10.1016/j.medj.2021.04.006> ↗

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Introduction to Git

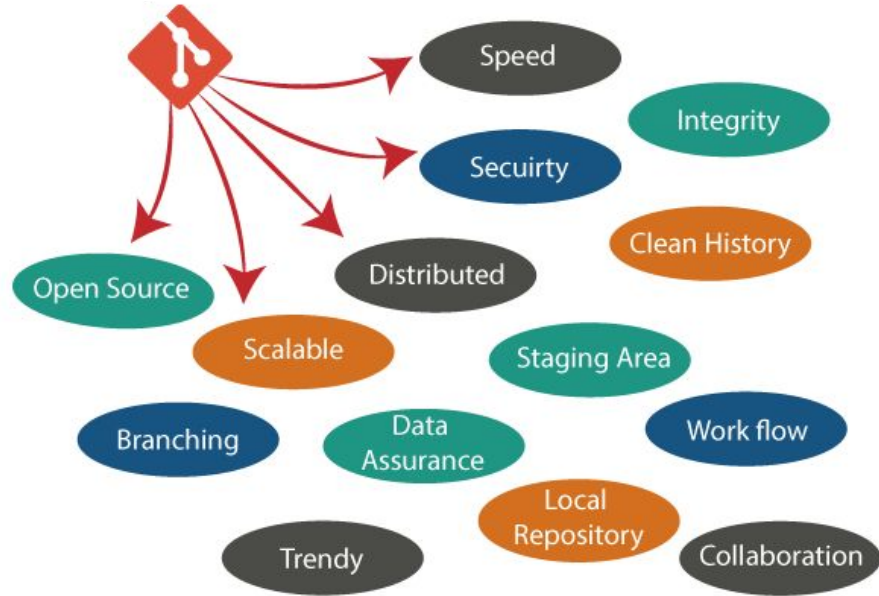
What is Git?

- **Version control system** that helps developers track and manage changes to code;
- Popular for **collaborative development** and maintaining a history of changes.



Why use Git?

- **Track Changes:** view a history of all code modifications;
- **Collaboration:** Allows multiple developers to work on a project simultaneously;
- **Branching and Merging:** Experiment with new features without affecting the main codebase;
- **Backup:** added protection against loss of work.



What is a Git Repository?

- A Git repository is the **.git/** folder;
- Any work done on a project is tracked in the **.git/** folder;
- The **.git/** folder is the **database** that contains all the information about the project:
 - builds **history** over time

What's inside the `.git/` directory?

<code>.git/</code>	→	The hidden <code>.git/</code> folder is created after running the 'git init' command.
<code>HEAD</code>	→	A label (ref) that points to the currently checked-out commit.
<code>config</code>	→	A file containing Git configuration settings for the local repo.
<code>description</code>	→	A file containing an optional description for your Git repo.
<code>hooks/</code>	→	A folder containing 'hooks' which allow triggering custom scripts after running Git actions.
<code>info/</code>	→	A folder containing the 'exclude' file, tells Git to ignore specific file patterns on your system.
<code>objects/</code>	→	A folder containing Git's object database, which stores the objects representing code files, changes and commits tracked by Git.
<code>refs/</code>	→	A folder holding the refs (labels) Git uses to represent branches & tags.

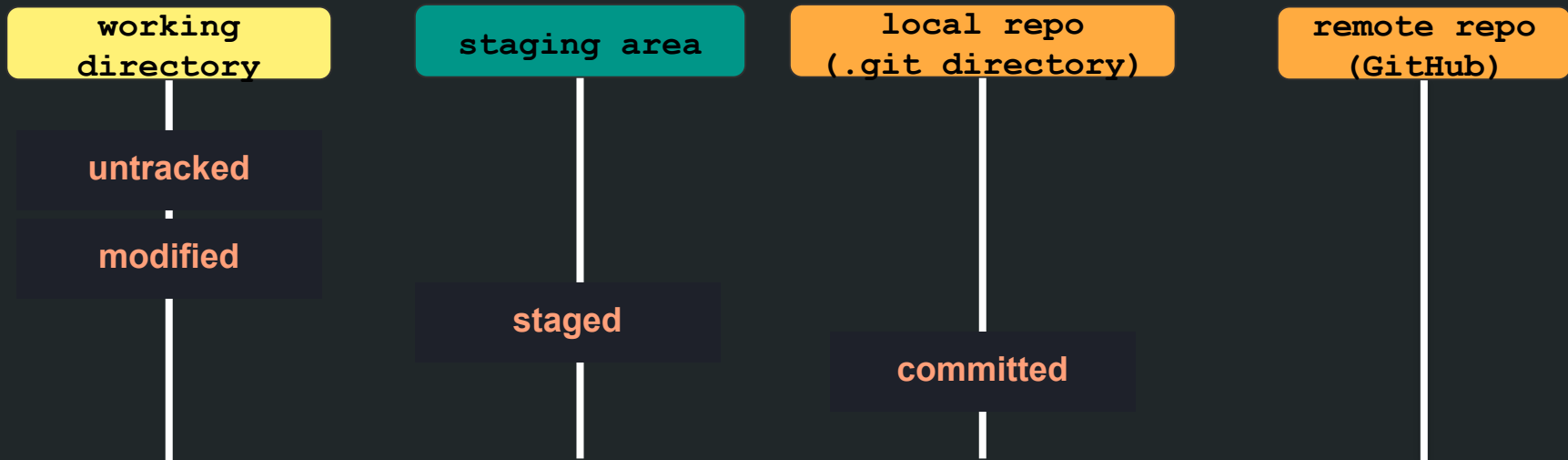
working
directory

staging area

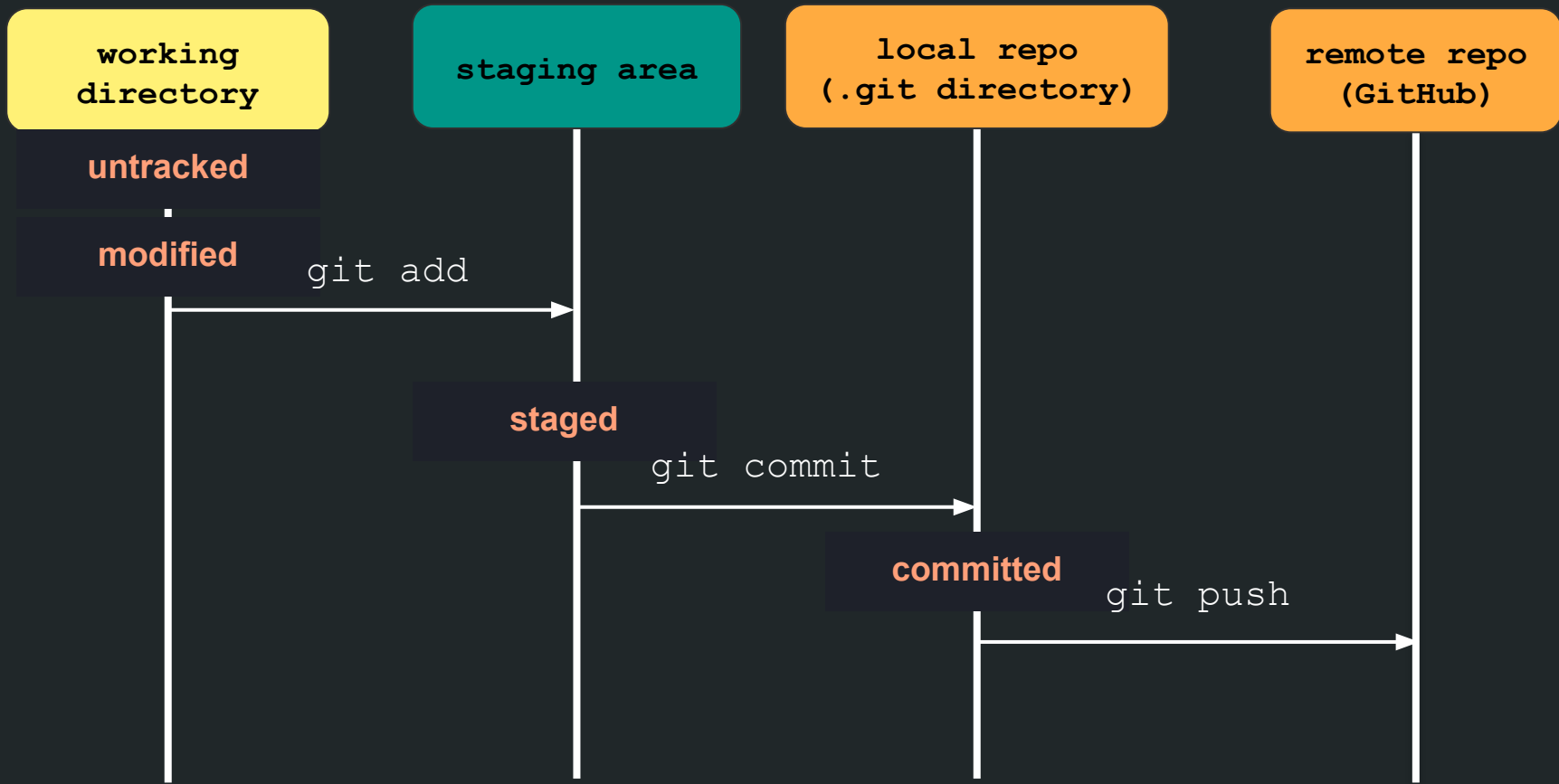
local repo
(.git directory)

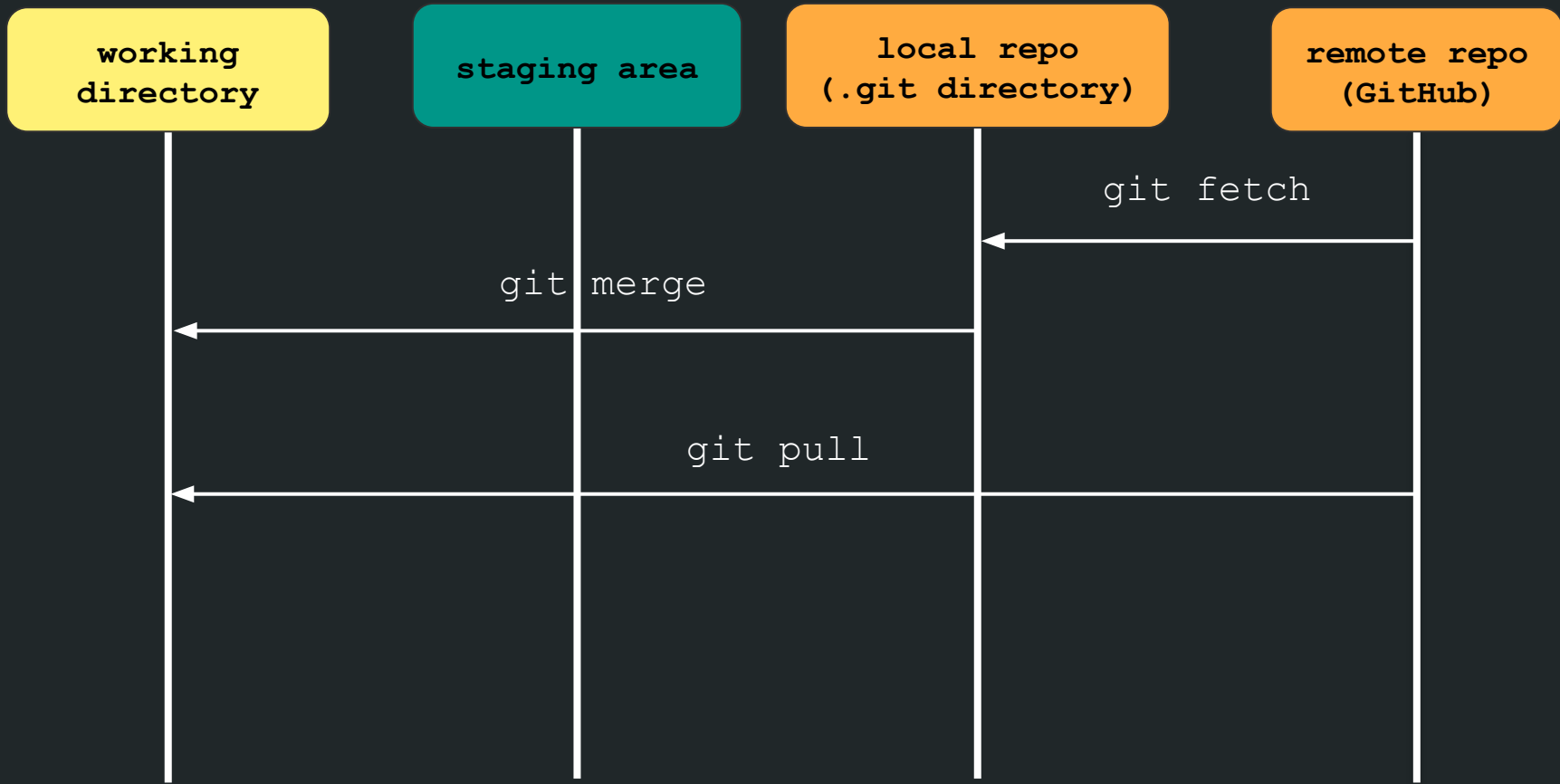
remote repo
(GitHub)

1. **Working Directory**: Where you make changes to your code.
2. **Staging Area**: Where you stage your changes for committing
3. **Repositories**: Where your commits are stored
 - can be **local (.git)** or **remote (GitHub)**



untracked	- file that has not been added to git
modified	- previously added file that has now been modified
staged	- file that will be part of the next commit
committed	- captured state in the commit history





Thoughts? Questions?
